

Understanding & Attacking Large Language Models

How LLMs work under the hood, how they are trained and augmented, what they are used for — and a threat-intelligence deep dive into the malicious "dark LLMs" sold to cybercriminals.

[How LLMs work](#)[GPT vs BERT](#)[Fine-tuning & RAG](#)[Use cases](#)[WormGPT · FraudGPT · XXXGPT](#)

01 Introduction to Large Language Models

Definition

A **Large Language Model (LLM)** is a machine-learning model trained on very large amounts of text so it can **predict and generate human-like language**. "Large" refers to two things at once: the size of the **training data** (often hundreds of billions of words / tokens) and the size of the model itself — its **parameters**, the learned numerical weights, ranging from hundreds of millions to hundreds of billions or more.

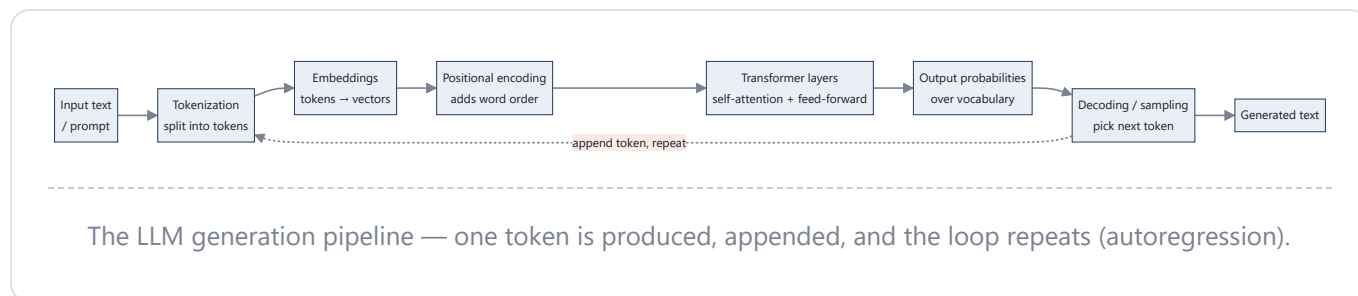
At its core an LLM is a **probability engine over language**. Given some input text, it estimates *which token is most likely to come next*, and repeats that prediction to build sentences, paragraphs, and code. It does not "look up" answers — it generates them statistically from patterns learned during training.

KEY TERM — TOKEN

The unit an LLM actually reads and writes. A token is roughly a word or word-fragment: `security` might be one token, while `unbelievable` could split into `un` + `believable`. Models operate on tokens, not raw characters.

How LLMs Work

Almost all modern LLMs are built on the **Transformer** architecture (from the 2017 paper "*Attention Is All You Need*"). Its defining innovation is the **self-attention mechanism**, which lets the model weigh how much every token should influence every other token — capturing context and long-range relationships far better than the older RNN / LSTM approaches.



Step by step:

1. **Tokenization** — the prompt is broken into tokens.
2. **Embeddings** — each token is mapped to a high-dimensional vector encoding meaning; similar concepts sit close together in this vector space.
3. **Positional encoding** — because attention has no inherent sense of order, position information is added so the model knows word sequence.
4. **Transformer layers** — a stack of blocks, each with **multi-head self-attention** (every token "attends to" others to gather context) and a **feed-forward network**. Stacking many layers builds increasingly abstract representations.
5. **Output layer** — produces a probability distribution over the whole vocabulary for the *next* token.
6. **Decoding / sampling** — a token is chosen. Parameters like **temperature** (randomness) and **top-p** / **top-k** (how many candidates to consider) control how deterministic or creative the output is.
7. **Autoregression** — the chosen token is appended and the process repeats, one token at a time, until the response is complete.

INTUITION — SELF-ATTENTION

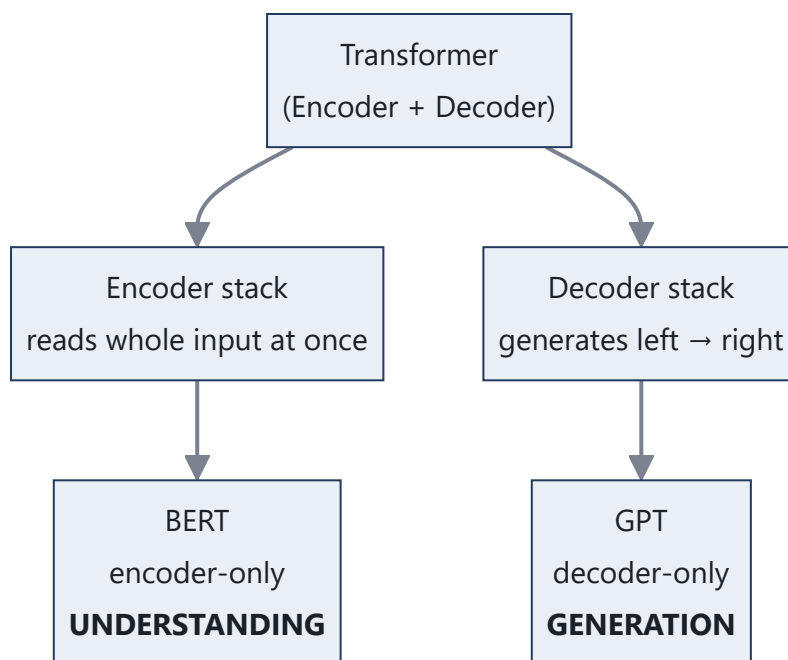
In "The analyst opened the alert because **it** looked suspicious," attention is what lets the model link "it" to "the alert" rather than "the analyst." Every token asks "which other tokens matter to me right now?" and weights them accordingly.

Importance & Impact in AI

- **A shift to general-purpose models.** Before LLMs, most NLP systems were narrow — one model per task. A single LLM performs *many* tasks with no retraining, just by changing the prompt.
- **Emergent capabilities.** As models scale, they show abilities not explicitly trained for — few-shot learning, step-by-step reasoning, code generation.
- **Natural language as the interface.** People interact with complex software in plain language, lowering the barrier to entry.
- **Foundation for the current AI wave.** LLMs underpin chat assistants, coding copilots, search, and autonomous "agents."
- **Security relevance.** The same accessibility is double-edged — LLMs *lower the barrier for attackers* (fluent phishing, malware scaffolding, reconnaissance at scale) and introduce a new attack surface (the model, its prompts, training data, and retrieval sources). That is exactly the shift that produced the malicious tooling in Section 5.

02 Understanding LLMs — GPT & BERT

GPT and BERT are both Transformer-based, but they use **different parts** of the Transformer and are optimized for **different jobs**. This split is the cleanest way to understand the model landscape.



BERT uses the encoder (understanding); GPT uses the decoder (generation).

GPT — Generative Pre-trained Transformer

- **Architecture:** decoder-only.
- **Directionality:** unidirectional (left-to-right) — it predicts the next token from everything before it and cannot "see the future" while generating. This is **causal** / **autoregressive** language modeling.
- **Training objective:** next-token prediction over massive text corpora.
- **Strength:** generating fluent, coherent, open-ended text.
- **Name unpacked:** *Generative* (produces new content) · *Pre-trained* (learns general language first from huge unlabeled data) · *Transformer* (the architecture).
- **Typical uses:** chatbots, writing assistance, summarization, code generation, Q&A.

BERT — Bidirectional Encoder Representations from Transformers

- **Architecture:** encoder-only.
- **Directionality:** bidirectional — reads the *entire* input at once, using context from **both left and right** of every token, for a richer understanding of meaning in context.
- **Training objectives:**
 - **Masked Language Modeling (MLM)** — random tokens are hidden and the model fills them in using surrounding context in both directions.
 - **Next Sentence Prediction (NSP)** — decide whether sentence B logically follows sentence A.
- **Strength:** understanding text — producing high-quality embeddings for downstream analysis. BERT does *not* generate long free-form text.
- **Typical uses:** classification, sentiment analysis, named-entity recognition, search relevance, semantic similarity.

Dimension	GPT	BERT
Transformer part	Decoder-only	Encoder-only
Context direction	Unidirectional (L→R)	Bidirectional (both ways)
Core objective	Next-token prediction	Masked LM + Next-Sentence Prediction
Primary talent	Generating text	Understanding text
Output	New free-form text	Embeddings / labels / spans
Best-fit tasks	Chat, writing, code, summarize	Classify, extract, search, similarity

MEMORY HOOK

GPT talks (generation), **BERT reads** (understanding).

03 Training & Augmenting LLMs

There are two ways to change what a model knows or how it behaves: **change the model** (foundational → fine-tuned) or **change what you feed it at run time** (RAG). They are often combined.

Foundational (Base) Model vs Fine-Tuned Model

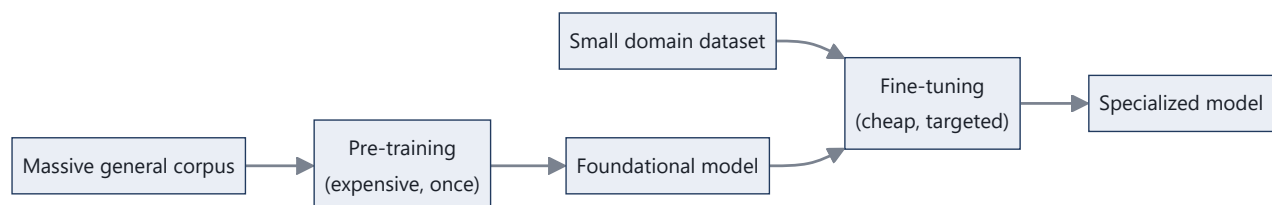
Foundational model — a.k.a. base / pre-trained model

- Trained once on an enormous, broad, general-purpose corpus.
- Extremely expensive and slow to produce (huge compute + data).
- Knows a little about almost everything, but is not specialized for any narrow task or organization.

- Examples of the *category*: base versions of the GPT family, LLaMA family, etc.

Fine-tuned model

- Takes an existing foundational model and trains it **further on a smaller, targeted dataset** to specialize it.
- Much cheaper than training from scratch — it reuses everything the base model already learned (a form of **transfer learning**).
- Adjusts behavior, tone, format, or domain expertise (legal, medical, security-log analysis, a company's writing style).
- Related — **instruction tuning** / **RLHF** (Reinforcement Learning from Human Feedback): fine-tuning steps that make a base model actually *follow instructions* and behave helpfully/safely, turning a raw text-predictor into an assistant.



Fine-tuning reuses the base model, so specialization is cheap compared with training from scratch.

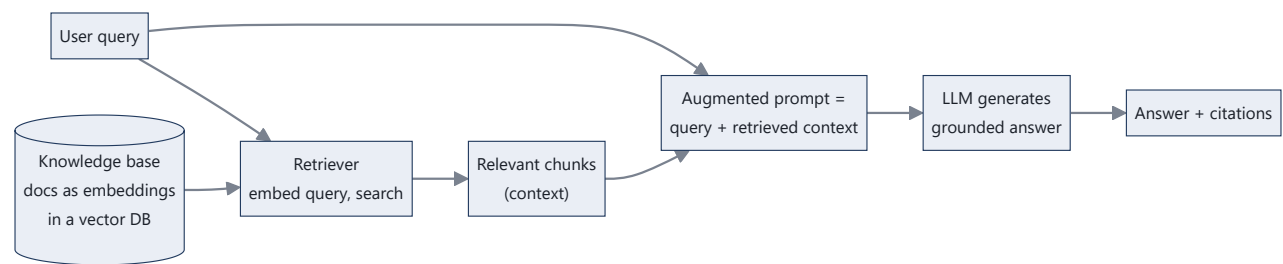
ANALOGY

The foundational model is a **university graduate** with broad general knowledge. Fine-tuning is the **on-the-job training** that turns them into a specialist for *your* team.

Retrieval Augmented Generation (RAG)

The problem RAG solves: a model's knowledge is **frozen at its training cutoff** and lives only inside its weights. It cannot know your private documents or today's news, and when it lacks a fact it may **hallucinate** — state something fluent but false.

The RAG idea: instead of retraining, **retrieve relevant, up-to-date information at query time** and insert it into the prompt as context, so the model generates an answer *grounded in* that retrieved material.



RAG grounds the model in fetched source text rather than relying only on frozen weights.

How it works:

1. Documents are split into chunks, converted to **embeddings** (vectors), and stored in a **vector database**.
2. At query time the user's question is also embedded and used to **search** the vector DB for the most semantically similar chunks (**semantic search**).
3. The retrieved chunks are **prepended to the prompt** as context.
4. The LLM answers using that supplied context, ideally citing its sources.

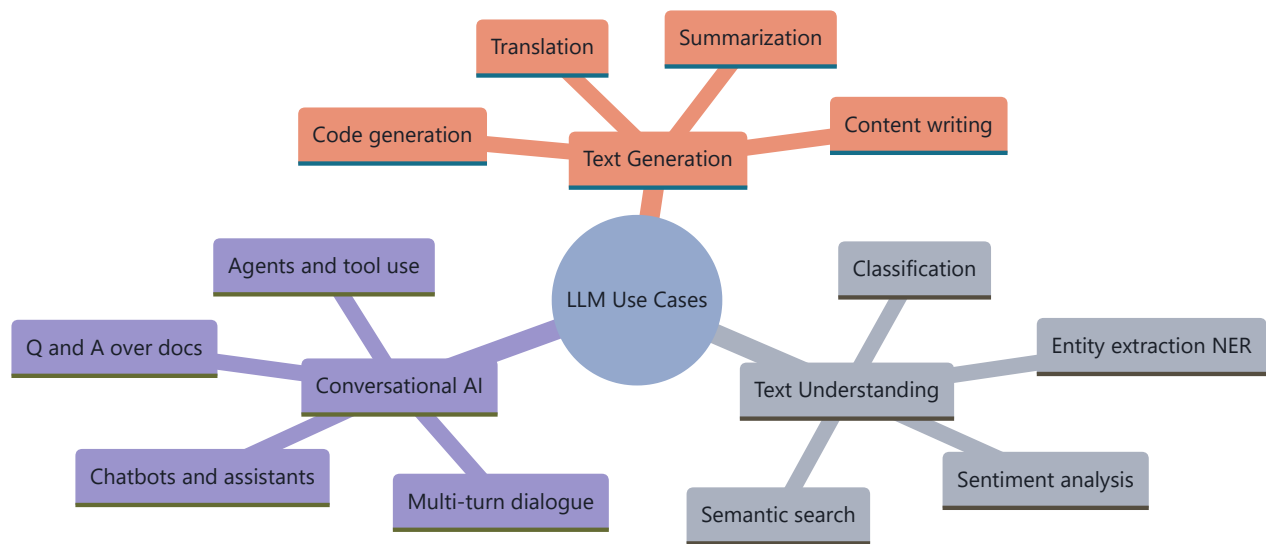
Why RAG matters: it adds current / private / proprietary knowledge without retraining, **reduces hallucination** by grounding answers in real source text, enables **citations** and traceability, and is cheaper to update than fine-tuning — you just update the knowledge base.

	Fine-tuning	RAG
Changes	The model's weights / behavior	The inputs at run time
Best for	Teaching <i>style, format, skills, behavior</i>	Supplying <i>facts & fresh / private knowledge</i>
Updating knowledge	Requires re-training	Just update the knowledge base
Hallucination	Doesn't inherently fix it	Directly reduces it via grounding

SECURITY NOTE

RAG introduces a fresh attack surface. If an attacker can poison the knowledge base or plant malicious text that gets retrieved, they can steer the model's output — known as **RAG poisoning / indirect prompt injection**.

04 Use Cases of LLMs



The three broad families of LLM application.

Text Generation

Producing **new** text from a prompt: drafting articles, emails, and reports; **summarizing** long documents; **translating** between languages; and **generating and explaining code**. This is GPT-style, generative territory.

Text Understanding

Analyzing and extracting meaning from existing text rather than producing prose: **classification** (spam / not-spam, topic labels), **sentiment analysis**, **named-entity recognition** (pulling out people, orgs, IOCs), and **semantic search** (matching by meaning, not keywords). This is BERT-style territory — directly relevant to SOC work like alert triage and phishing classification.

Conversational AI

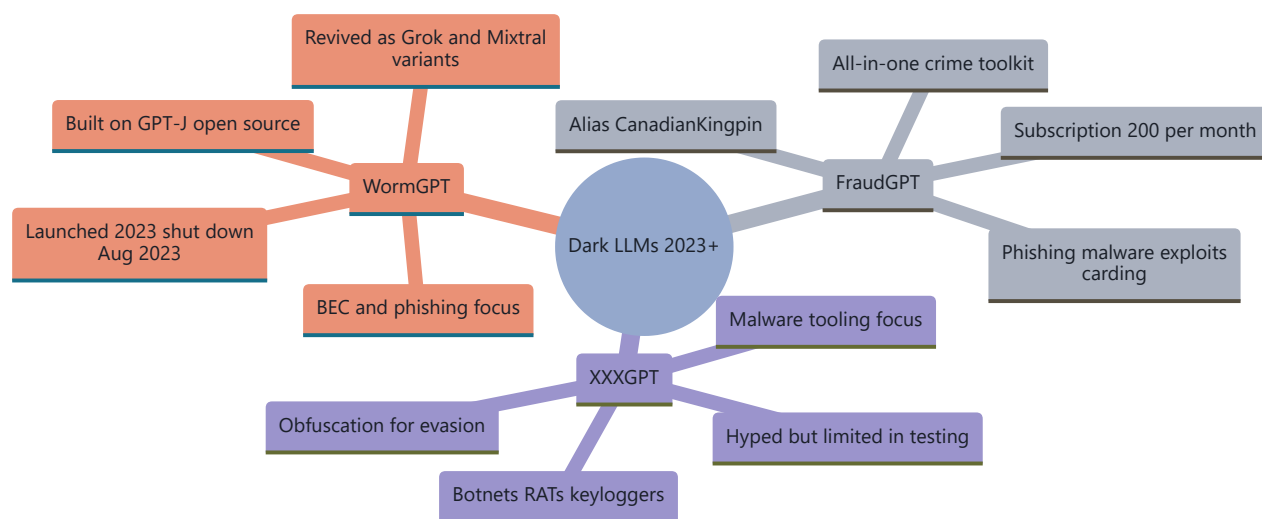
Combining generation and understanding into **multi-turn dialogue**: chatbots and virtual assistants, **question answering over documents** (often RAG-powered), and increasingly **agents** that call tools/APIs and take actions on a user's behalf. Conversational context — memory of the dialogue so far — is what makes these systems feel coherent, and what makes them manipulable.

05 Real-World Malicious LLM Tools

By mid-2023, dark-web forums developed a market for "**dark LLMs**" — models stripped of safety guardrails (purpose-built, fine-tuned without safety constraints, or "jailbroken" wrappers around legitimate models) and sold, often on a **subscription / crime-as-a-service** basis. Their significance is less about raw power and more about **lowering the barrier to entry**: unskilled actors gain fluent phishing, malware scaffolding, and reconnaissance help, letting attacks **scale**.

TWO WAYS A "DARK LLM" IS BUILT

- 1 — **Purpose-built / fine-tuned**: an open-source base model (e.g. GPT-J) trained further on malware- and fraud-related data with no safety layer.
- 2 — **Jailbroken wrapper**: a thin front-end over a legitimate commercial or open model, using a manipulated *system prompt* to bypass its guardrails. Much cheaper to run — and, as researchers later found, what several "brands" actually were.



The three tools on this syllabus, and their distinct focuses.

WormGPT

The first commercialized dark LLM — the "evil twin of ChatGPT"

STATUS: original shut down Aug 2023 · brand revived 2024–25

FIRST REPORTED	July 2023 by SlashNext (development teased on Hack Forums in March 2023; public release June 2023)
ATTRIBUTED TO	A developer using the alias " Last ", later reported to be a young Portuguese programmer (named in press as Rafael Morais)
BASE MODEL	The open-source GPT-J (EleutherAI, ~6 billion parameters, released 2021)
PRIMARY FOCUS	Business Email Compromise (BEC) & phishing; also malware code snippets
SOLD ON / PRICING	Hack Forums; reported around €60–100 / month (~\$110), with a much pricier private tier cited in some reports
DISTINCTIVE TRAIT	No ethical guardrails; strong at fluent, grammatically clean, multilingual lures

What it was

WormGPT was one of the **first widely recognized commercialized malicious LLMs**, built by fine-tuning the open-source GPT-J model and removing the safety constraints found in mainstream tools like ChatGPT and Google Gemini. Its headline capability was generating **persuasive BEC and phishing emails** at scale — fluent, free of the tell-tale grammar mistakes that once helped people spot scams, and easily produced in multiple languages. It was also advertised for generating malicious **code snippets** (malware scaffolding), letting less-skilled actors assemble and modify malware faster.

Why it mattered

WormGPT proved the "jailbroken / uncensored AI" business model worked and could be sold as a subscription — the formal arrival of LLM capability inside the existing **crime-as-a-service** ecosystem. A single operator could now generate hundreds of unique, polished lures in the time it once took to write one, which directly **defeats email defenses that rely on spotting repeated patterns** or bad grammar.

Its end — and its afterlife

Heavy media attention led the original creator to **shut the project down on 8 August 2023**. But the shutdown did not end the story: "**WormGPT**" **became a brand**. Between late 2024 and early 2025, security researchers at Cato CTRL found new WormGPT-branded services on BreachForums, sold via Telegram chatbots. Crucially, these were **not bespoke models** — they were **jailbroken wrappers over legitimate commercial LLMs**: one variant (`xzin0vich-WormGPT` , Oct 2024) ran on **Mistral's Mixtral**, and another (`keanu-WormGPT` , Feb 2025) ran on **xAI's Grok**, both steered by manipulated system prompts to bypass the models' guardrails. This is the clearest illustration of the whole category's trajectory: adapting an existing API is far easier than training a malicious model from scratch.

FraudGPT

The "all-in-one" cybercriminal toolkit (a.k.a. FraudBot)

STATUS: emerged July 2023 · subscription service

FIRST REPORTED	Circulating since ~ 22 July 2023 ; exposed by Netenrich's threat-research team (researcher Rakesh Krishnan)
ATTRIBUTED TO	Alias " CanadianKingpin " — a previously established dark-web-marketplace vendor who moved to Telegram to avoid marketplace "exit scams"
BASE MODEL	Undisclosed ; the seller claimed it could leverage "different AI models"
PRIMARY FOCUS	Broad fraud toolkit: spear-phishing, scam pages, malware/exploits, carding, cracking, vuln/leak discovery
PRICING	~ \$200 / month , ~\$1,000 / 6 months, or ~\$1,700 / year; first buyers offered a free month; "24/7 escrow"
CLAIMED TRACTION	Seller claimed 3,000+ sales and reviews by end of July 2023

What it was

Where WormGPT concentrated on email lures, FraudGPT was marketed as a **one-stop offensive toolkit "with no boundaries."** Its advertised capabilities spanned writing convincing **phishing / spear-phishing emails** and social-engineering content, building **phishing and scam pages**, generating **malware and exploit code** (including "undetectable" malware), **carding** and **cracking** tools, and **finding leaks, vulnerabilities, and compromised credentials** — plus general hacking guidance. In one demo captured by Netenrich, it produced a short, professional-looking bank-themed phishing message on request.

The business model

FraudGPT is the clearest example of **weaponized AI as a productized service**: a tiered subscription, promotional claims of thousands of sales, an escrow system, promised fortnightly updates, and a "privacy focus" pitch. Its whole value proposition was **democratization** — handing inexperienced attackers a starting skill set that previously required real effort to build.

[Link to WormGPT](#)

Netenrich assessed that the actor behind FraudGPT was **likely the same group operating WormGPT** — consistent with a small cluster of operators driving the early dark-LLM market rather than many independent developers.

XXXGPT

The malware-tooling specialist — botnets, RATs, keyloggers

STATUS: announced July 2023 · efficacy disputed

ANNOUNCED	July 2023 on the XSS cybercrime forum, with promotional posters and fanfare
POSITIONING	"Revolutionary" bot AI customization with " no censorship or restrictions "; claimed to be backed by a team of five
PRICING	Reported around \$90 / month
PRIMARY FOCUS	Malware tooling : botnets, Remote Access Trojans (RATs), keyloggers, crypters, info/cryptostealers, ATM malware kits
SELLING POINT	An obfuscation feature to help camouflage generated code and evade detection
REALITY CHECK	Testers suspected a jailbroken prompt wrapper ; some demo output was still censored, and code samples were simple and easily caught by antivirus

What it claimed to be

XXXGPT pushed the dark-LLM concept **beyond phishing into offensive malware development**. It was advertised as generating code for **botnets** (for large-scale, automated attacks and bot-driven fraud), **RATs** (giving attackers remote control of a victim's device for data theft or manipulation), **keyloggers** (capturing passwords and card details from keystrokes), plus crypters, stealers, and ATM-malware kits. It leaned on an **obfuscation** capability, pitched as making its output harder for defenders to detect.

The reality — a useful skepticism lesson

XXXGPT is a good reminder that **dark-market marketing runs well ahead of capability**. Independent researchers who examined it were skeptical: one common criticism on the forums was that the "product" looked like little more than a **jailbroken prompt**, and testers of the demo found it **still returned censored responses** in places. The code it did generate (e.g. a keylogger sample) tended to be **simple and readily flagged by most antivirus**. The

genuinely alarming part was not the sophistication of the output but the fact that such code could be requested and generated **on demand, at scale, with no skill barrier**.

Side-by-Side Comparison

	WormGPT	FraudGPT	XXXGPT
Emerged	Mid-2023	July 2023	July 2023
Signature focus	BEC & phishing lures	All-in-one fraud toolkit	Malware: botnets, RATs, keyloggers
Base model	GPT-J (known)	Undisclosed	Undisclosed / likely wrapper
Pricing	~€60–100 / mo	~\$200 / mo → \$1,700 / yr	~\$90 / mo
Operator alias	"Last"	"CanadianKingpin"	Team of five (XSS forum)
Reputation	Genuine capability; brand later revived	Aggressively marketed toolkit	Hyped; efficacy questioned

Defensive Takeaways

- **Retire the "spot the typo" heuristic.** AI-written lures are fluent and multilingual; user-awareness training must shift toward *context, intent, and verification of requests* (especially payment / credential changes) rather than spelling and grammar.
- **Lean on controls that don't depend on message content.** Email authentication (SPF / DKIM / DMARC), attachment / link detonation, identity and MFA hardening, network segmentation, and **UEBA / behavioral analytics** limit the blast radius when a convincing lure lands.
- **Track the market.** This ecosystem moves fast — monitor threat-intel feeds, ISAC sharing, and advisories from bodies like CISA and the FBI; expect new brands, rebrands, and jailbroken-wrapper services.

- **Assume scale, not genius.** These tools mostly *lower the barrier to entry* — so plan for *more*, more-personalized, and more-frequent attacks rather than fundamentally novel ones.

BIG-PICTURE (2025–2026)

Threat intelligence increasingly notes that packaged, paid dark-LLMs have become **less relevant**: attackers now self-host **open-source models** (fine-tunable without safety constraints) behind friendly UIs, or **jailbreak legitimate models** — often for equal or better results at lower cost. WormGPT, FraudGPT, and XXXGPT were the vanguard of a much larger ongoing shift toward **the industrialization of cybercrime with AI**.

FRAMING

These tools are covered here for **threat awareness and defense** — understanding what adversaries have, how they market it, and what it means for your controls. The value for a defender is in detection, user training, and control design.

06 Quick Revision Sheet

Concept	One-liner
LLM	Transformer-based model that predicts the next token to generate / understand language
Self-attention	Lets each token weigh the relevance of every other token → context
Autoregression	Generate one token, append, repeat
GPT	Decoder-only, unidirectional, generation
BERT	Encoder-only, bidirectional, understanding
Foundational model	Broad, expensive, trained once on general data
Fine-tuning	Cheap specialization of a base model on targeted data
RAG	Retrieve fresh / private facts at query time → ground the answer, cut hallucination
WormGPT	2023, GPT-J based, BEC / phishing; shut down Aug 2023, brand revived on Grok & Mixtral
FraudGPT	2023, all-in-one crime toolkit; "CanadianKingpin"; ~\$200/mo, 3,000+ claimed sales
XXXGPT	2023, malware / RAT / botnet focus; ~\$90/mo; hyped but limited in testing
Core threat theme	Dark LLMs lower the barrier to entry and let attacks scale

07 References

Factual claims about the malicious tools were checked against the following threat-intelligence reporting. Foundational LLM / Transformer concepts are standard material (the 2017 "Attention Is All You Need" paper is the primary architectural source).

Malicious LLM tooling

Netenrich — *FraudGPT: The Villain Avatar of ChatGPT* (original disclosure) — netenrich.com

The Hacker News — *New AI Tool 'FraudGPT' Emerges* — thehackernews.com

Infosecurity Europe — *The Dark Side of Generative AI: Five Malicious LLMs* — infosecurityeurope.com

Cato Networks (Cato CTRL) — *WormGPT Variants Powered by Grok and Mixtral* — catonetworks.com

TechRepublic — *Two WormGPT Clones That Use Grok and Mixtral* — techrepublic.com

Huntress — *What Is WormGPT?* — huntress.com

Palo Alto Unit 42 — *The Dual-Use Dilemma of AI: Malicious LLMs* — unit42.paloaltonetworks.com

Sophos — *Cybercriminals can't agree on GPTs (XXXGPT skepticism)* — sophos.com

Fingerprint — *LLM-generated fraud: malware & vulnerabilities (XXXGPT)* — fingerprint.com

Kaspersky DFI — *Shadowy innovation: cybercriminals experiment with AI* — dfi.kaspersky.com

Study notes prepared for defensive-security learning. The dark-LLM landscape changes quickly — verify tool status and pricing against current threat-intelligence reporting before relying on specifics.